How Much of Your VLA is kNN

Denis Grachev[1] **Damien LaRocque**[1] **Omar Moured**[1,2] [1]Sereact GmbH [2]Karlsruher Institut für Technologie

CoRL 2026 Workshop on The Action Gap: Large Generative Models Learn Rich Knowledge

91.7 % — LIBERO success with retrieval alone, no training

−9.3 %pt — smallest drop under perturbation of any policy

7 of 8 — backbones retrieve better after fine-tuning

4 of 9 — VLAs lose to *k*-NN on their own backbone

MOTIVATION

What does a benchmark score tell us?

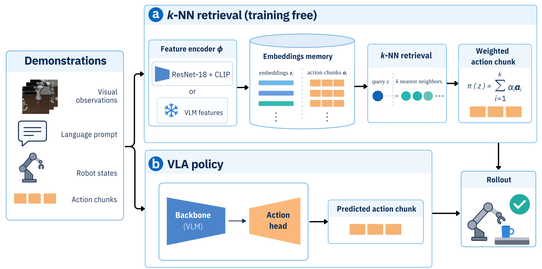
VLAs now score above 95 % on LIBERO. But a success rate on its own says little: it depends on how closely the test conditions resemble the demonstrations.

We test how far **retrieval alone** gets, then use it to analyze nine public VLAs:

1. How much of LIBERO can retrieval solve?
2. What does fine-tuning change in the backbone and the action head?
3. Can we score a retrieval key without running rollouts?

METHOD

k-NN as a training-free policy



1. **Memory:** frames from the 1,693 LIBERO demonstrations, each stored with the action chunk that follows it.
2. **Retrieve** the k nearest frames and execute a weighted average of their action chunks:

$$\pi(q) = \sum_{i \in \mathcal{N}_k(q)} \alpha_i \, \mathbf{a}_{t_i:t_i+H}, \qquad \alpha_i \propto \exp\big(-d(q,i)/\tau\big)$$

3. **Swap in a VLA's VLM backbone** as the encoder to evaluate it separately from the action head.

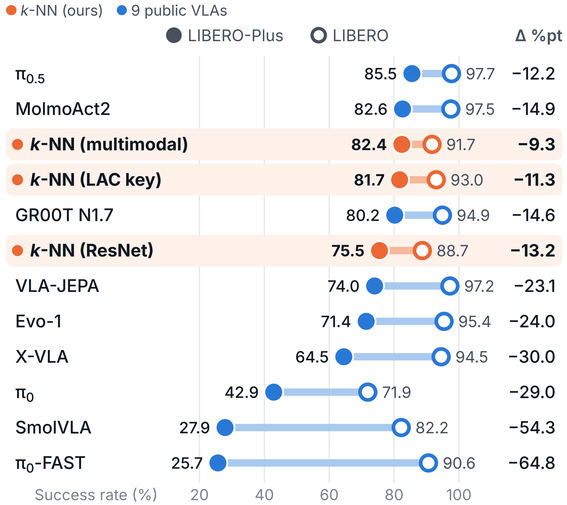
_k_-NN (ResNet) — **88.7 %**
Frozen ResNet-18 image features, searching only the instructed task.

_k_-NN (multimodal) — **91.7 %**
Adds robot state, language, time and history. Searches all demonstrations.

_k_-NN (LAC key) — **93.0 %**
Multimodal, plus a ResNet projection trained with LAC.

RESULT 1

Retrieval reaches 91.7 % on LIBERO and drops the least under perturbation



Success rate (%). LIBERO: 2,000 episodes. LIBERO-Plus: the same 420 variants.

- With no learned policy, *k*-NN **matches π_0-FAST** on LIBERO.
- On LIBERO-Plus, the multimodal *k*-NN **beats 7 of the 9 VLAs**.

ABLATION

What does retrieval need?

LIBERO success (%) after removing or replacing one component:

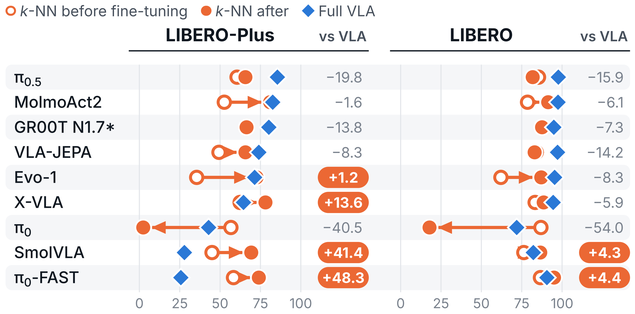
***k*-NN (multimodal)**	**91.7**
Without mode seeking, history, state or time (one at a time)	**90.7–91.5**
DINOv2-S features instead of ResNet-18	**83.4**
Only 10 % of the demonstrations	**77.4**
No language instruction	**64.6**
Raw pixels instead of image embeddings	**57.2**

The demonstrations, the image embeddings and the language prompt matter most.

RESULT 2

Retrieval separates the VLM backbone from the action head

We swap the ResNet features for each VLA's own VLM features, before and after LIBERO fine-tuning.



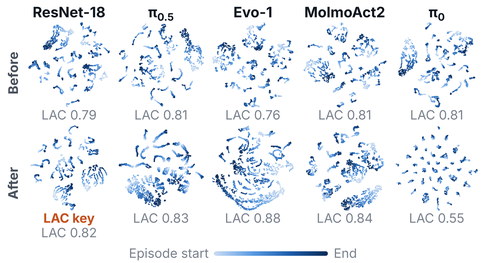
Success rate (%). vs VLA: *k*-NN after fine-tuning minus the full VLA, in %pt. *Frozen VLM.

- **When *k*-NN beats the full VLA** (by up to 48 %pt), the action head is leaving useful information in the backbone unused.

RESULT 3

What LIBERO fine-tuning does to the backbone

Why can *k*-NN on a backbone outperform the full VLA? Fine-tuning reshapes the backbone:



t-SNE of the retrieval keys for 4,150 demonstration frames, colored by episode progress. Below each map: LAC@8.

- **Better retrieval under perturbation** for 7 of 8 fine-tuned VLAs.
- **Less scene-dependent:** nearest neighbors also become closer in task progress for most models.
- **One exception, π_0:** representations collapse into task clusters, and *k*-NN success drops from 86.9 % to 17.9 %.

RESULT 4

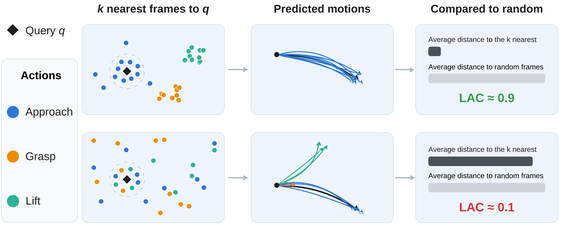
VLA actions stay close to the demonstrations

We compare each VLA's action chunks from its LIBERO rollouts with the training demonstrations. Held-out demonstrations serve as the baseline.

- *k*-NN predicts the actions of **5 of 8 VLAs** better than it predicts held-out demonstrations.
- **All 8 VLAs** are closer to a training action chunk than held-out demonstrations are.

METRIC

Local Action Consistency (LAC)

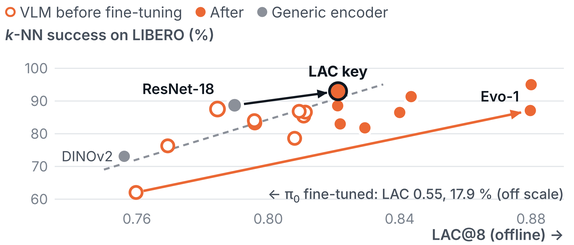


An **offline metric**: do a frame's k nearest neighbors $N_k(q)$ also have similar future action chunks, compared with other frames $C(q)$ from the same task?

$$\mathrm{LAC@}k(q) = 1 - \frac{\bar{d}_{\mathcal{N}_k(q)}}{\bar{d}_{\mathcal{C}(q)}}, \qquad \bar{d}_{\mathcal{S}} = \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} \|\mathbf{a}_q - \mathbf{a}_i\|_2^2$$

RESULT 5

LAC tracks retrieval success



Each point is one visual encoder. Retrieval uses its image features only, within the instructed task.

- **Correlates with closed-loop *k*-NN success** (Pearson r = 0.78).
- **Trains a better key:** ResNet *k*-NN rises from 88.7 to 93.0 %.

Report a retrieval baseline alongside VLA results to distinguish memorization from generalization.

Limited to LIBERO and LIBERO-Plus, where dense demonstrations may favor retrieval. Retrieval cannot go beyond what is in its memory.

Paper & videos

Code